

Power BI vs Real-Time Dashboards

Not just about the looks

 <https://brianbonk.dk>
 <https://aka.bi/slides>
 <https://aka.bi/brian>
 <https://kql.how>

Brian Bønck

 ProBI.dk



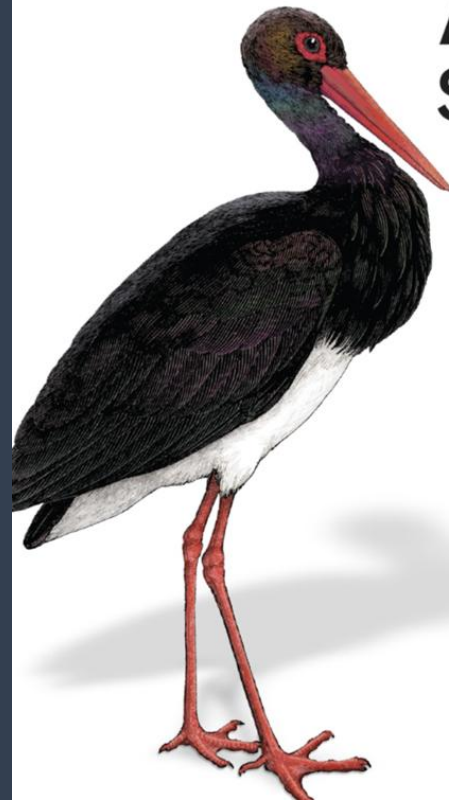
FastTrack Recognized
Solution Architect

Certified trainer



O'REILLY®

Microsoft Fabric Analytics Engineer Associate Study Guide



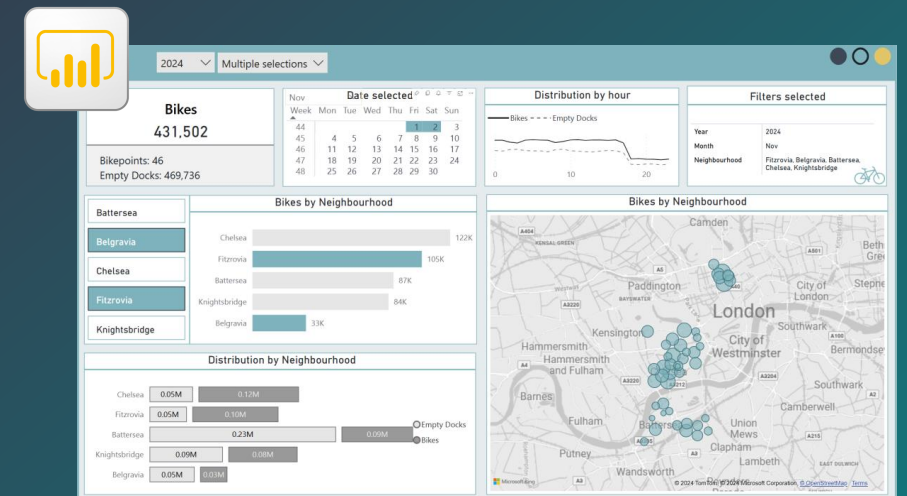
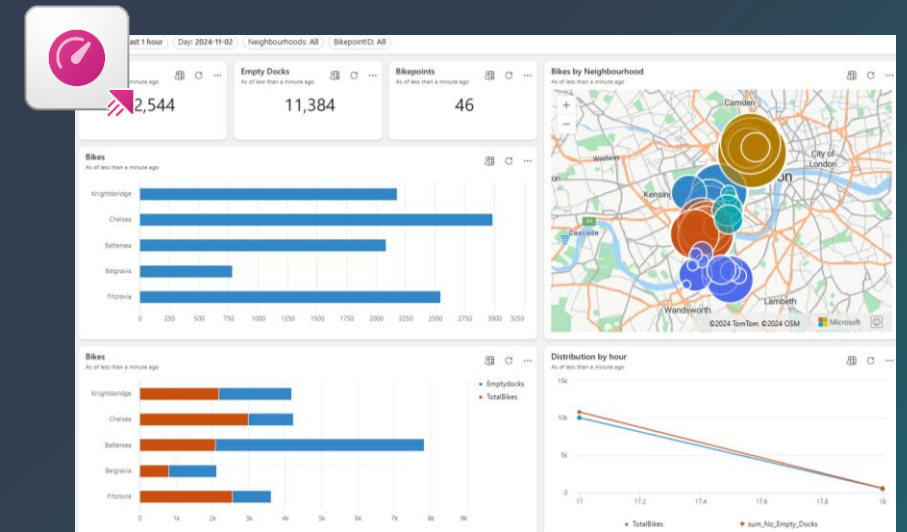
Hands-on Practice
and Expert Tips for
Acing the DP-600
Certification Exam

Brian Bønck & Valerie Junk



Real-Time vs Power BI

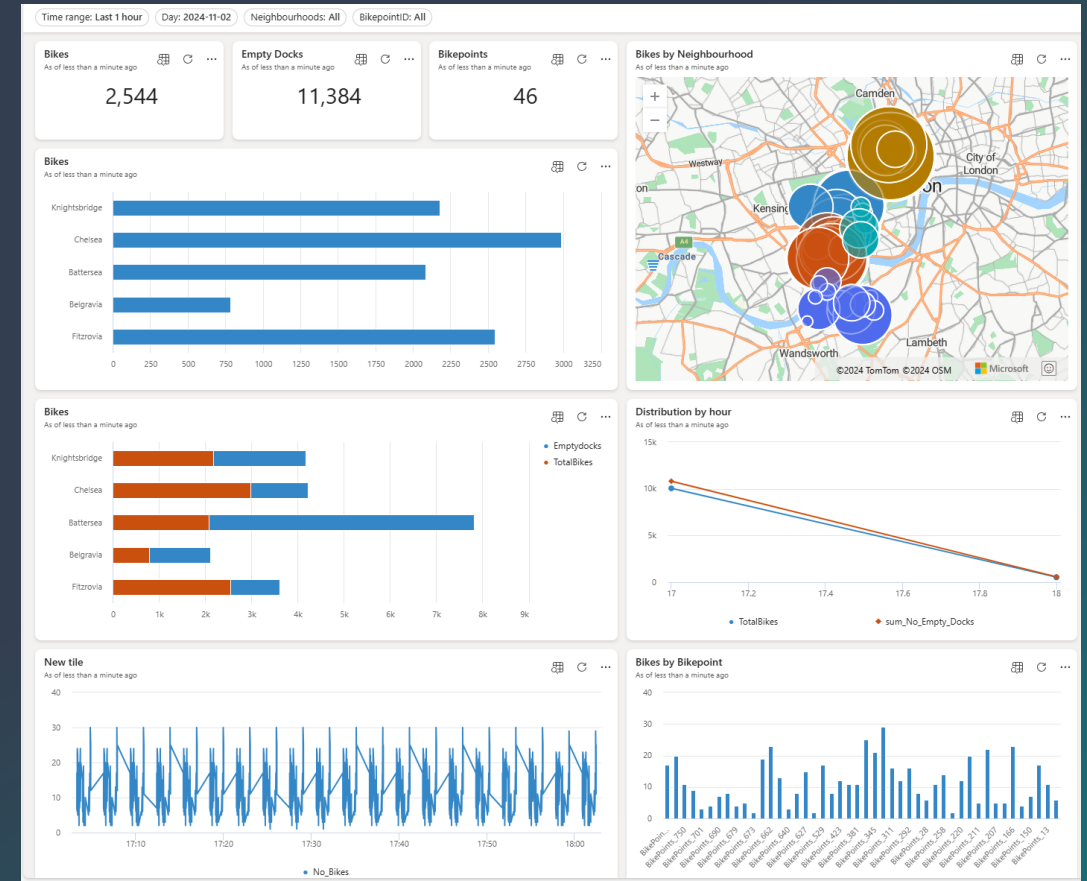
- **Power BI** and **Real-Time Dashboards** are both great options, each designed for different types of data needs and scenarios.
- Power BI: Built for deep insights, complex transformations, and interactive reporting.
- Real-Time Dashboards in Fabric: Ideal for event-driven, near real-time monitoring with immediate data updates and alerting.





Real-Time Dashboard

- “Always on” – continuous data tracking
- High-frequency data updates
- Near real-time alerting
- Licensing: Fabric Capacity
- Perfect for monitoring any scenario with constantly updating data.



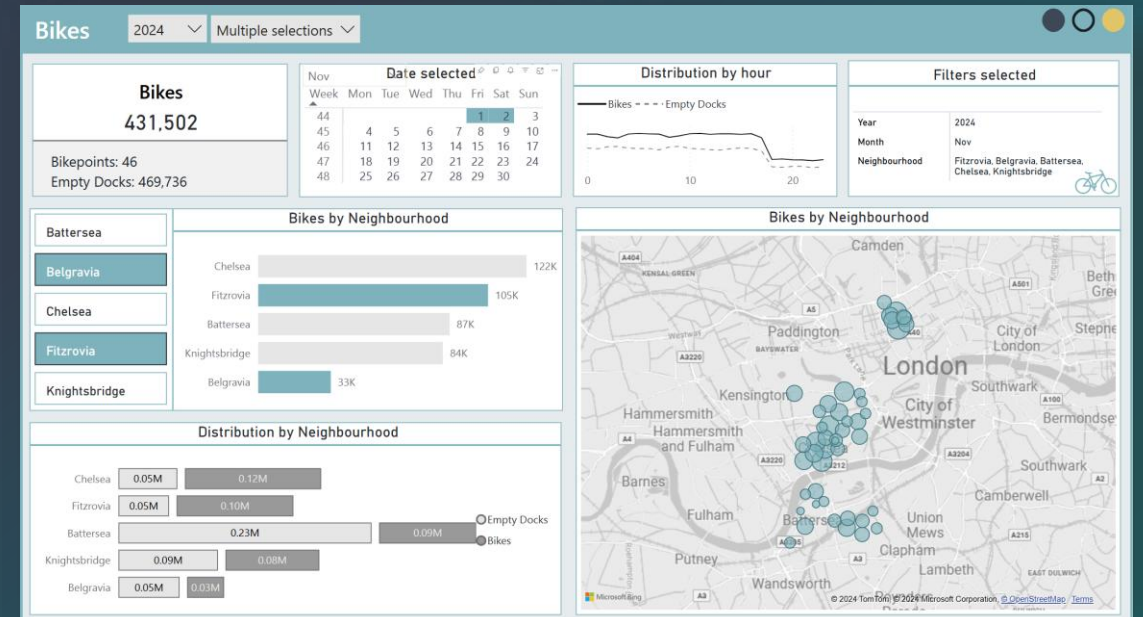
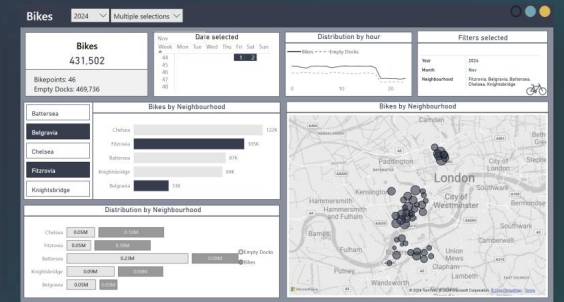
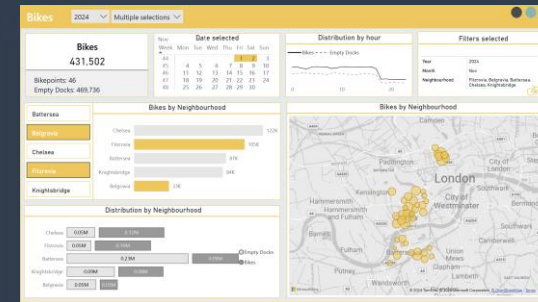
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Power BI Report

- User experience (tailored insights)
- Easy Interaction & filter selection
- DAX
- Data visualization

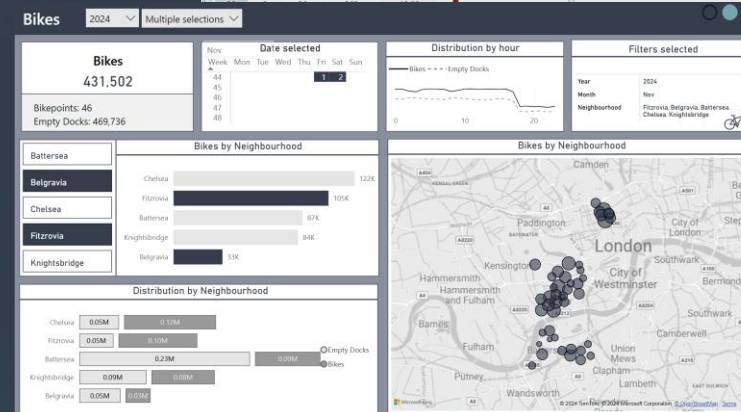
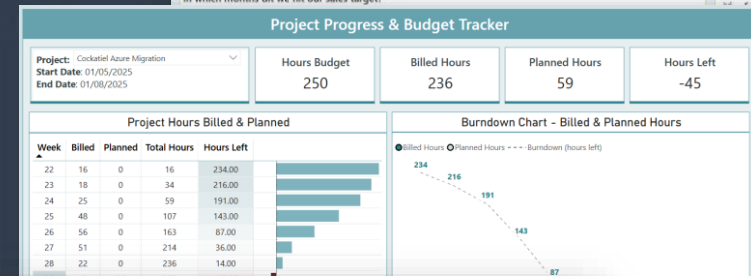
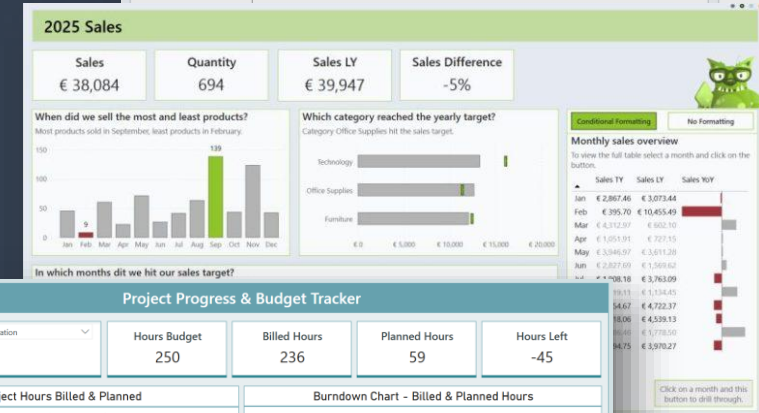
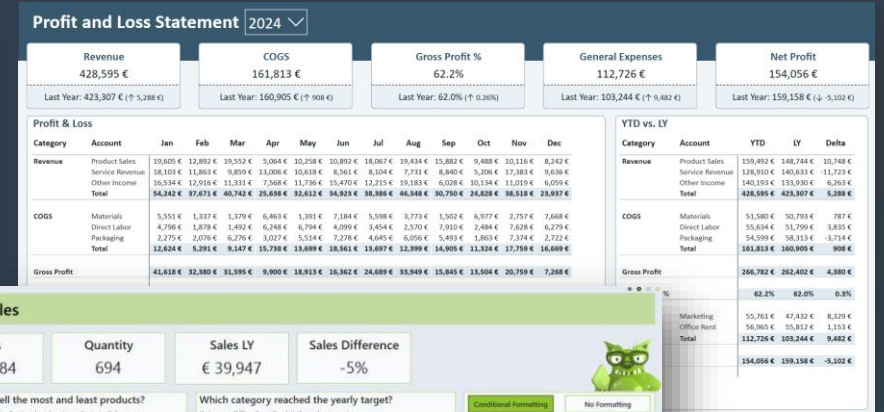
Licensing: Pro, PPU, E5 or F64+





Use Cases

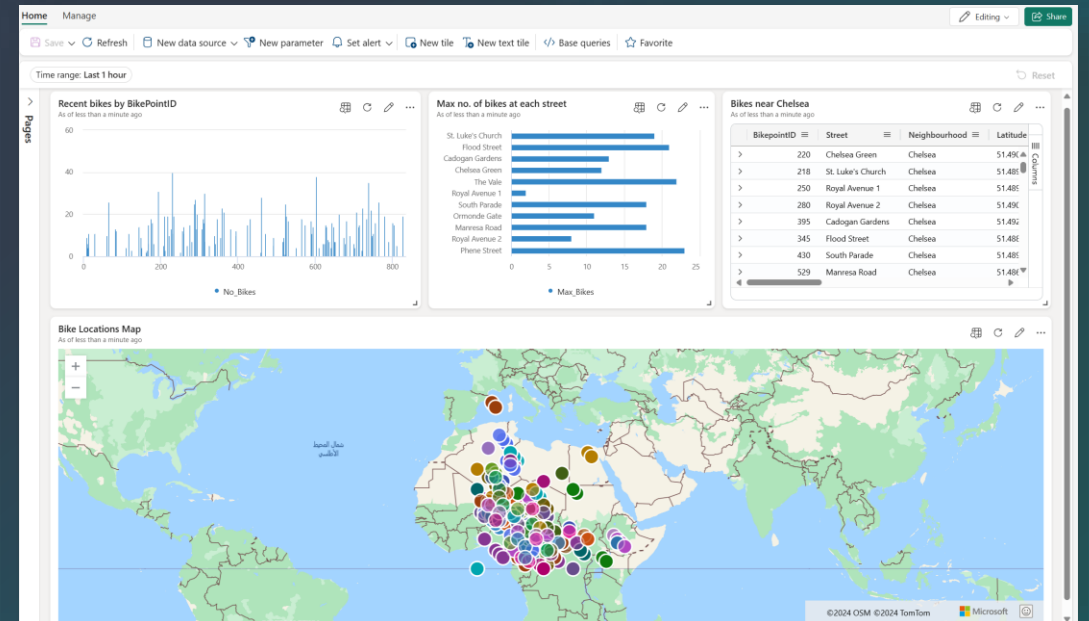
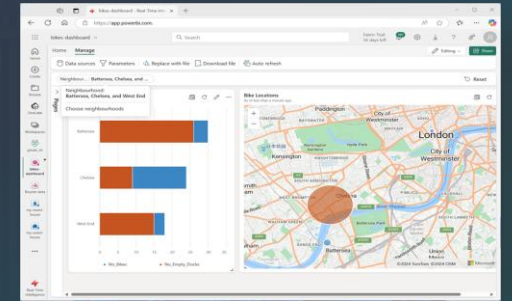
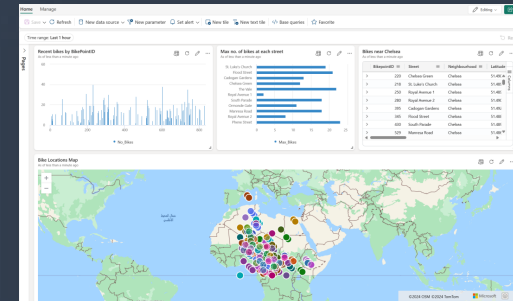
- (In depth) Data Analysis
 - Financial Analysis
 - Sales & Marketing
 - Human Resource Analysis
 - Supply Chain Optimization
 - Compliance
- What if Analysis
- Patterns & Trends





Real-Time Dashboard

- Visuals build for speed
- Manual interaction and filtering
- KQL
- Data insights and actions



Free – only consumes capacity units



Use Cases

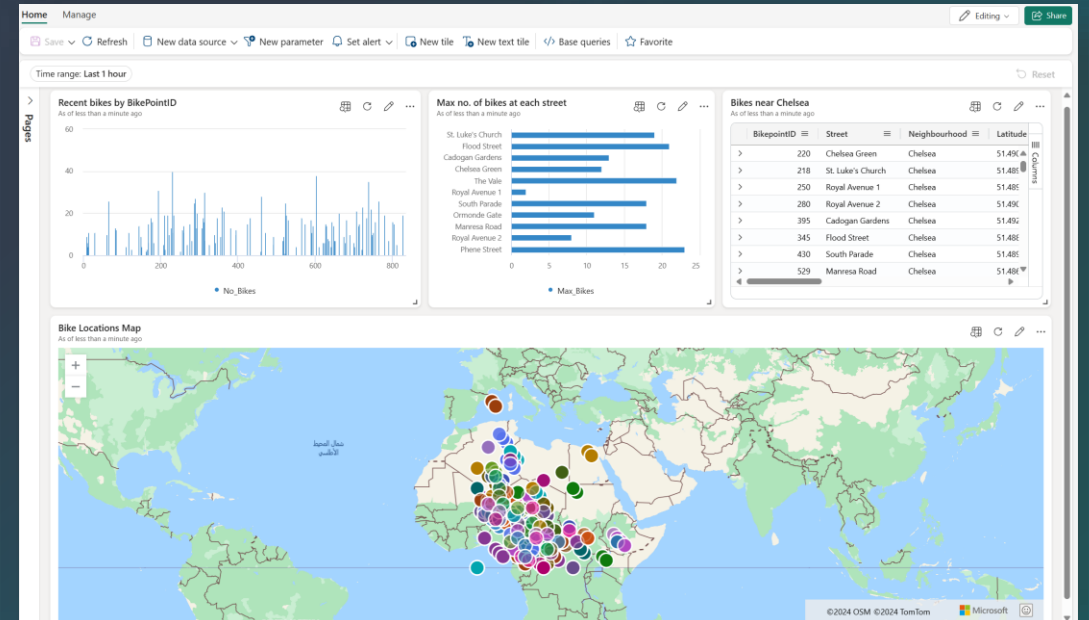
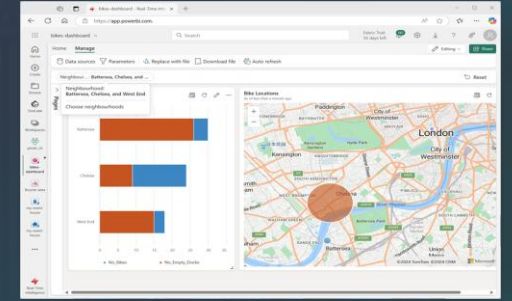
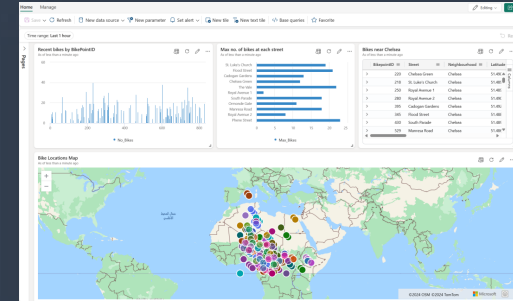
- Short time to insights
 - Immediate status of trends and data points
 - Near real-time analytics
 - Outlier detection and action
 - Graph analysis
 - Telco, windmills, racing etc.
- Historical trends
- Windowing





User Experience

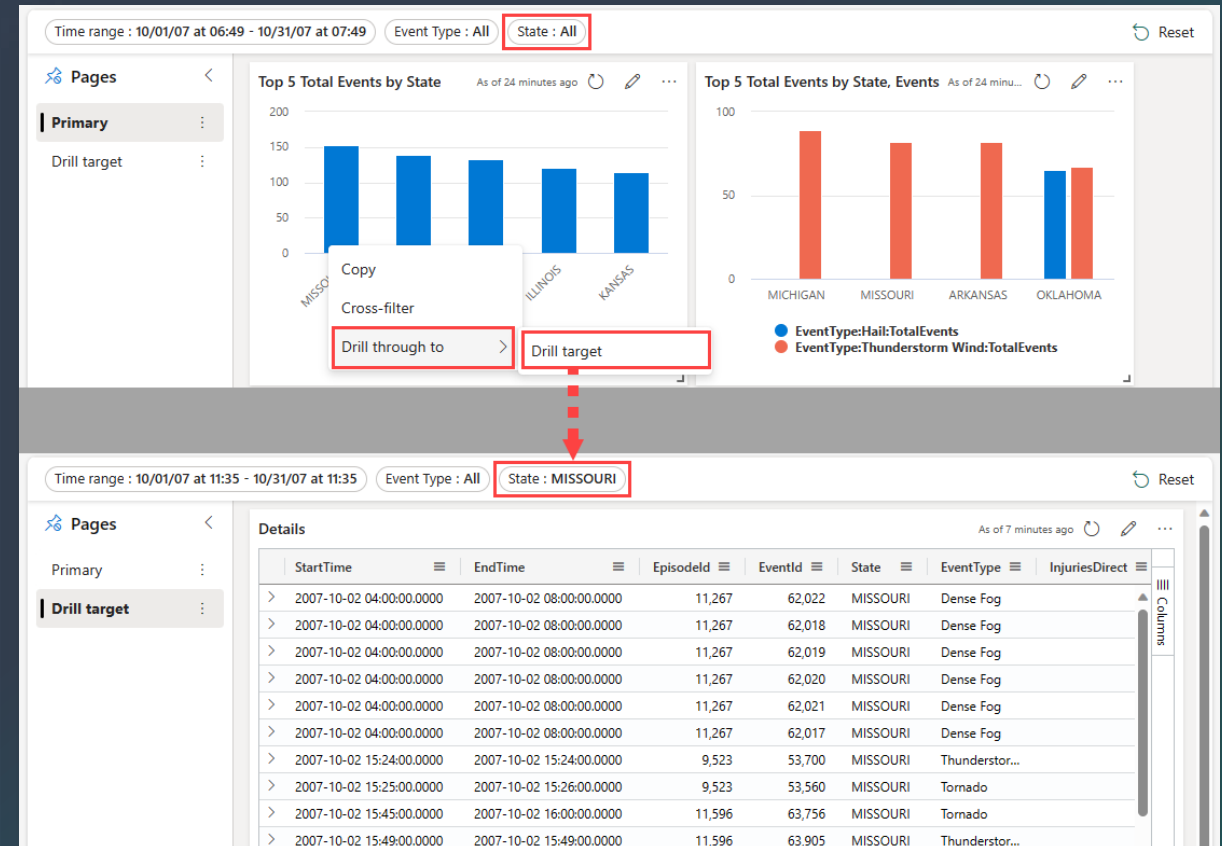
- Fixed reporting with fixed layouts
- Limited set of standard visuals
- Plotly extension for Python based reporting
- Built-in visuals at query time
- Cross filtering must be done manually





Interaction & filter selection

- All filters must be created manually
- Extensive date/time filtering
- Drill through options





KQL (& Dataviz)

- Datasets somewhat reusable
- Limited set of standard visuals
- Some conditional formatting (colours and icons)

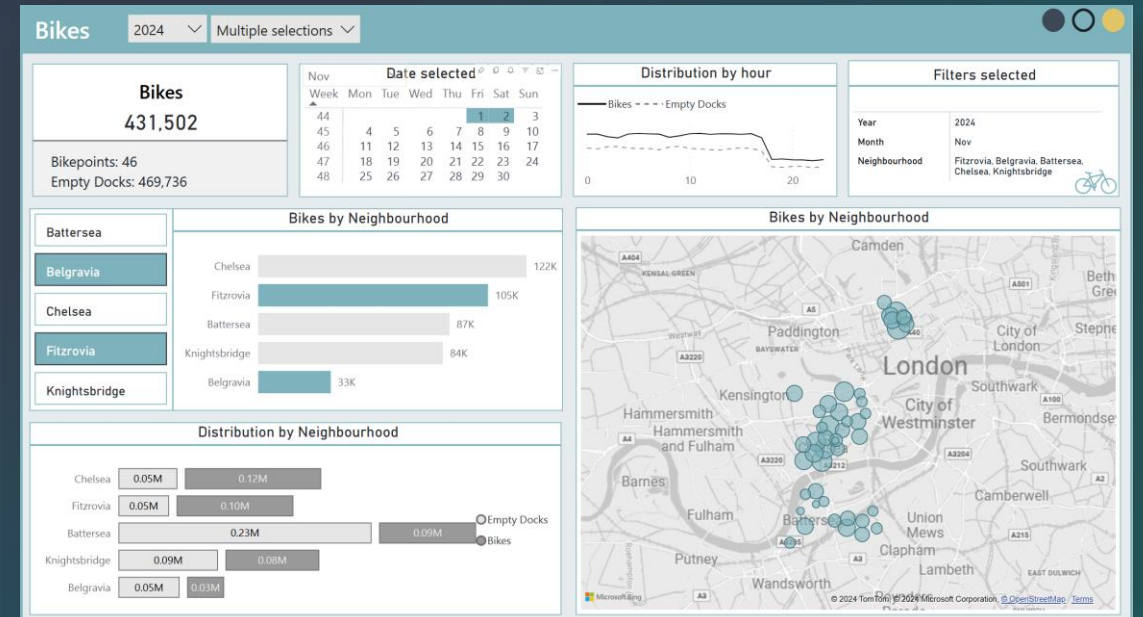
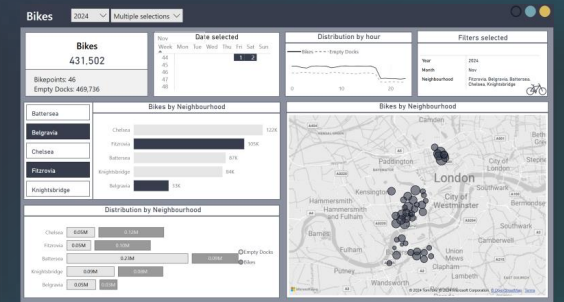
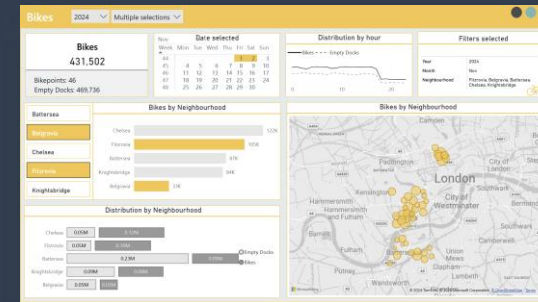
Cold/Wind Chill ⚠ 17	Excessive Heat 6	Flash Flood 1
Heat ⚠ 20	Heavy Rain 1	High Wind 1
Marine Strong Wind 1	Thunderstorm Wind 7	Tornado ⚠ 31

EventType	Count
> Cold/Wind Chill	17
> Excessive Heat	6
> Flash Flood	1
> Hail	1
> Heat	20
> Heavy Rain	1
> High Wind	1
> Hurricane (Typhoon)	1
> Marine Strong Wind	1
> Thunderstorm Wind	7
> Tornado	31
> Wildfire	8
> Winter Weather	2



User Experience

- Personalize reports
- Visuals can be created for a specific audience
- Easy to use and interactive





- | Neighbourhood | BikePoint | Street | Bikes | Empty Docks | Time |
|---------------|-----------|--------------|-------|-------------|----------|
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:06:34 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:09:38 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:12:42 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:15:47 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:18:51 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:21:55 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:25:00 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:28:04 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:31:08 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:34:12 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:37:16 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:40:20 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:43:24 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 09:46:29 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:38:45 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:41:49 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:44:54 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:47:58 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:51:02 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:54:07 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 10:57:11 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:00:15 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:03:19 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:06:23 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:09:27 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:12:31 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:15:37 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:18:42 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:21:46 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:24:50 |
| Knightsbridge | 662 | Phene Street | 24 | 0 | 11:27:54 |
| Total | | | 39331 | 17006 | |

Bikes by Neighbourhood

Bikes by Street

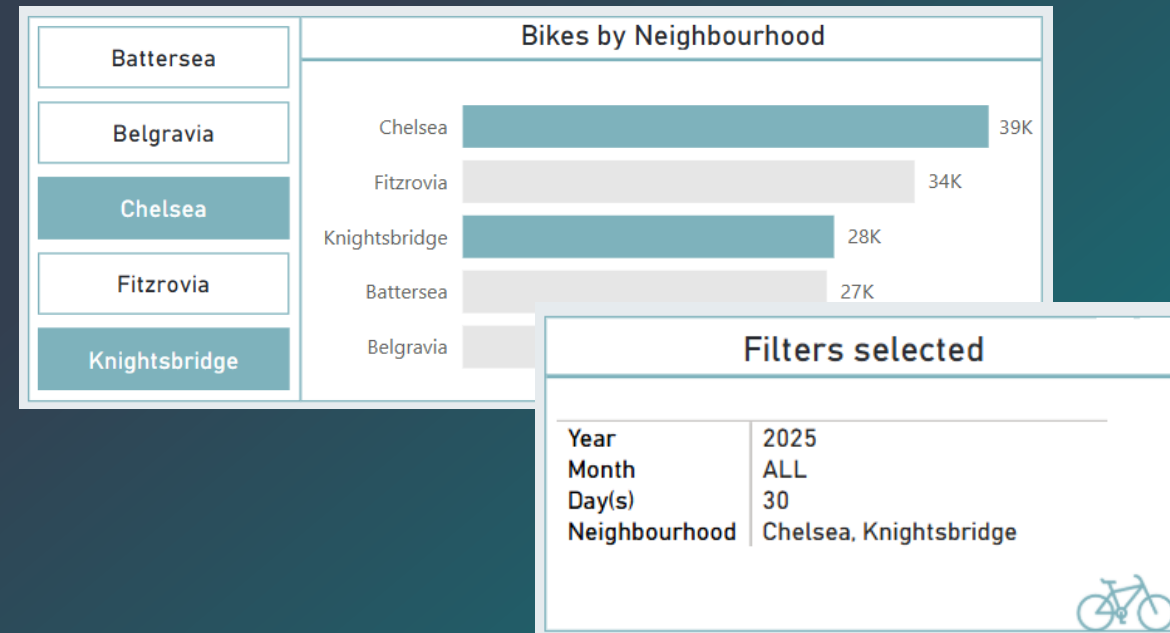
Empty Docks

Empty Docks



DAX (& Dataviz)

- DAX measures (reusable)
- Wide range of native visuals (bar, line, pie, KPI, maps).
- Conditional formatting for action driven storytelling.



Feature comparison

Real-Time vs Power BI

Feature	Real-Time Dashboards	Power BI
Refresh Rate	Near real-time (always on)	Scheduled refresh
Interactivity	Limited interactive	Interactivity
Visualization	Limited	Highly customizable
Alerting	Teams, email or Fabric item	Teams & email
Use Case	Event data, traffic, IoT	In depth & action-driven

POWER BI vs REAL-TIME DASHBOARDS



FILTERING



COST



VISUALIZATION



DATA ACCESS



SHARING



REAL-TIME
DASHBOARD

DYNAMIC TIME
RANGE
FILTERING

THE
EVENTHOUSE
COST IS THE
ONLY CU USAGE

LIMITED
GALLERY. CAN
USE PLOTLY FOR
CUSTOM VIZ

ONLY KQL DATA
AND IN "DIRECT
QUERY"

SHARING
REQUIRES
ACCESS TO THE
KQL DATA



POWER BI
REPORT

EASY FILTER ON
EVERYTHING

Power BI
Licences

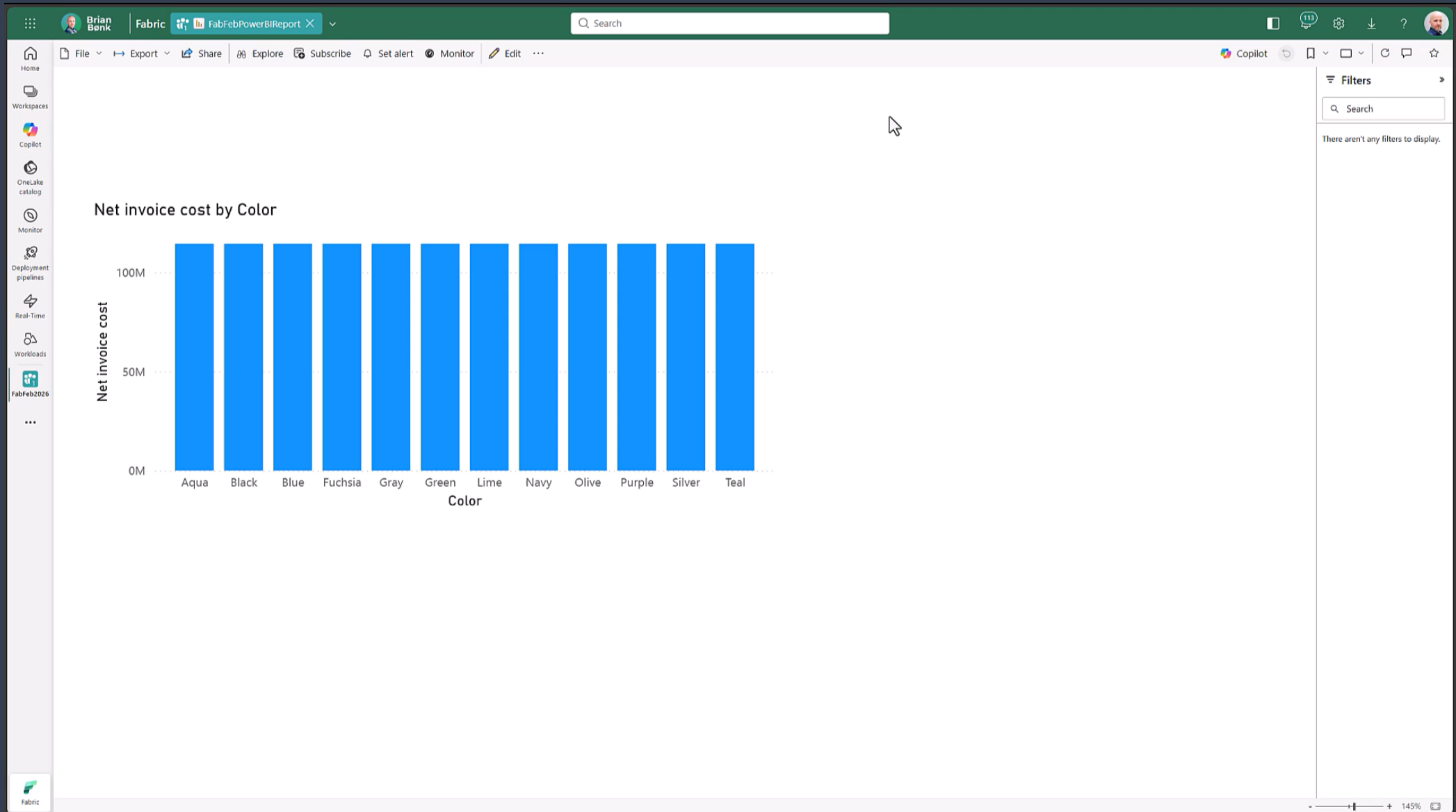
RICH GALLERY
WITH EXTENSIVE
CUSTOMIZATION
CAPABILITIES

IMPORT, DIRECT
QUERY, DIRECT
LAKE

EASY SHARING
BOTH INTERNAL
AND EXTERNAL

The two tools are complimenting each other and have different purposes

POWER BI vs REAL-TIME DASHBOARDS



When use what?

**When use
what?**

Question	Tool
• Real-time updates?	• Real-Time Dashboards
• In-depth exploration?	• Power BI
• Alerting critical?	• Real-Time Dashboards
• Complex, multi-layered report?	• Power BI

Try it yourself!

- 8-step tutorial with sample data
- Create a Real-time dashboard
- Import results in Power BI



<https://learn.microsoft.com/fabric/real-time-intelligence/tutorial-introduction>

Takeaway

Real-Time Dashboards

- Always on, constantly changing
- Events, IoT sensors, Traffic
- High-frequency data updates



Power BI

- Strategic and Detailed Analysis
- In-depth data exploration with advanced visualizations and filtering



https://kql.how

 **kql.how**

- ▶ Queries
- ▶ Concepts
- ▶ APIs
- ▶ Management

The KQL language

KQL (Kusto Query Language) is a query language developed by Microsoft, specifically designed for querying large datasets, typically in cloud-based environments like Azure. It originated from a service called Kusto, which was initially developed for the internal needs of Microsoft for handling massive amounts of telemetry data, logging, and diagnostic data. Over time, it evolved into a more generalized query language, applicable to a variety of data analytics applications in the cloud.

KQL is a specialized query language tailored to meet the demands of querying large, real-time datasets, particularly in the realms of telemetry, monitoring, and log analytics. Originating from the internal Kusto engine at Microsoft, KQL has evolved to become a key language in the Azure ecosystem, allowing users to perform powerful data analysis and diagnostics across a variety of use cases. Its design reflects a blend of performance, scalability, and ease of use, making it an essential tool for modern cloud-based data environments.

The sections



Queries

This section contains the documentation for the data selection and manipulation.

[Read more](#)



Management

Here you will find the documentation for the management part of KQL.

[Read more](#)



API

The KQL language also has API capabilities, find the documentation here.

[Read more](#)

About this Reference

This KQL reference contains information on the KQL language. The Guide is updated regularly to reflect the latest changes in Microsoft products. Each documentation part is presented with a syntax, supporting articles and often with examples.

While this KQL Guide is a ProBI project created and maintained by [Brian Bönk](#), it also incorporates official Microsoft

<https://kql.how>

[illegible]

Thank you

